| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 12/08/2010 |
| **PI Name:** | Blancaflor, Elison B. Ph.D. | | |
| **Project Title:** | Actin Regulation of Arabidopsis Root Growth and Orientation During Spaceflight | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | elison.b.blancaflor@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | GOVERNMENT | **Phone:** | 321-501-5368 |
| **Organization Name:** | NASA Kennedy Space Center | | |
| **PI Address 1:** | Utilization and Life Science Office | | |
| **PI Address 2:** | Exploration and Research Technology Programs | | |
| **PI Web Page:** | | | |
| **City:** | Kennedy Space Center | **State:** | FL |
| **Zip Code:** | 32955 | **Congressional District:** | 8 |
| **Comments:** | NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma. | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2009 Space Life Sciences BRIC NNH09ZTT004N |
| **Start Date:** | 02/08/2010 | **End Date:** | 02/07/2012 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | Shuttle/ISS | | |
| **Flight Assignment:** | STS-131 | | |
| **Key Personnel Changes/Previous PI:** | None | | |
| **COI Name (Institution):** | Nakashima, Jin ( Noble Foundation )<br>Tang, Yuhong ( Noble Foundation ) | | |
| **Grant/Contract No.:** | NNX10AF43G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | We conducted experiments utilizing the BRIC-PDFU onboard STS-131 to understand how the actin cytoskeleton dictates root growth orientation during spaceflight. Our guiding hypothesis was that actin filaments (F-actin) negatively regulate environmental and endogenous signals to specify root orientation on earth and in space. This hypothesis was based on previous NASA-funded ground based research demonstrating that F-actin disruption enhanced the sensitivity of roots to gravity. We utilized Arabidopsis seedlings with a transfer (T)- DNA mutation in a vegetative actin isoform (act2) and the short-duration microgravity environment provided by the STS-131 mission to test our hypothesis. Microgravity was an essential component of our proposed experiments since it allowed us to determine the impact of other signals on root orientation that are typically masked by the strong gravitational force on earth. Using the two unpowered BRIC-LED units allotted to each investigator team, we quantified root orientation of dark-grown act2 and wild-type seedlings (aim 1). In parallel, we conducted an extensive set of genome-wide microarray studies using Affymetrix ATH1 arrays to unravel actin-dependent gene regulatory networks that modulate root growth and orientation during spaceflight (aim 2). Although such experiments required minimal manipulation from the flight crew, we were able to collect robust data that are paving the way for an in depth understanding of plant growth under microgravity, a goal consistent with the overall mission of NASA. Another significant deliverable of our project will be the inclusion of a photo-documentary article that will outline the activities of the selected Arabidopsis BRIC16 proposals in a book that will be published by Ikaros, a Japanese-based aviation magazine distributor. This will provide NASA with a new venue to highlight the significance of its life sciences space program to the general public. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research will lead to a better understanding of how plant development is affected by microgravity. Gravity dictates a number of important plant physiological processes here on earth that impact agricultural productivity. For example, gravity influences the downward growth of roots and upward growth of shoots (gravitropism), which are critical to optimize nutrient, water and light acquisition. Through a better understanding of the cellular and molecular processes underlying plant gravitropism, crops with improved growth habits and enhanced architecture can be developed here on earth. |
| **Task Progress:** | The major goal of this proposal is to determine how the actin cytoskeleton regulates the orientation of root growth and plant development in general under microgravity conditions.<br>The two aims in support of this goal were: 1). Quantify root orientation and determine amyloplast position in the root columella of Arabidopsis seedlings of wild-type and act2 mutants grown on the Biological Research in Canisters (BRIC)-Petri Dish Fixation Unit (PDFU) and 2). Compare transcript profiles of wild-type and act2 mutants (has a mutation in the vegetative actin 2 isoform) using Affymetrix ATH1 and GeneChip® Tiling 1.0R arrays to unravel actin-dependent gene regulatory networks that modulate seedling orientation during spaceflight. We have made significant progress with both aims.<br><br>In April 5, 2010, 14 60 mm Petri dishes containing Arabidopsis seeds were loaded into BRIC-PDFUs and launched to space on the Discovery STS-131 mission. Seeds germinated in space and grew in the dark for two weeks. Seedlings were returned to the KSC-SLSL after fixation in orbit by the crew of Discovery. To date, we have extracted RNA from seedlings and have completed ATH microarrays (aim 2). We have also completed the embedding of our seedlings on resin and started sectioning tissues for microscopic studies (aim 1).<br><br>The data we have obtained from the BRIC16 experiments are very robust. By including the act2 mutant in our experiments, we were able to include a strong positive control for our microarray work. Furthermore, the RNA integrity number (RIN), which is an indicator of the quality of RNA for microarray was very good. We found more than 830 genes affected in flight, with 694 down-regulated. 18% of down-regulated genes in space were involved in membrane function. The differential regulation of some of these genes have already been verified by quantitative RT-PCR (qRT-PCR).<br><br>A number of genes down-regulated in space were related to root function based on in silico expression analysis. Closer examination of root orientation during spaceflight showed a predominant left-handed coiling. Interestingly, a mutant to a gene that was 30 fold down regulated during spaceflight, displayed skewing of their roots on the ground, which roughly mirrored the behavior of wild type seedlings in microgravity.<br><br>In regard to aim 1, we found that seed germination and root length were not affected by the spaceflight environment. Preliminary ultrastructural analysis using transmission electron microscopy (TEM) revealed that act2 mutants in space have wavy cell walls compared to wild type. It is worth noting that roots of act2 mutants tend to wave more strongly on the ground when grown on an inclined hard agar surface. We are investigating whether the altered ultrastructure of act2 mutants in space can explain why roots of act2 mutants have an increased tendency to wave.<br><br>Overall, we have obtained a robust set of spaceflight microarray data that has now been verified by q RT-PCR and mutant analysis. The use of act2 mutant served as an excellent positive control for which to gauge the reliability of our microarray data set. During spaceflight, roots exhibited a pronounced coiling response and knockouts to a gene that was most strongly down regulated during spaceflight displayed a skewing response on the ground, which was consistent with the coiling of roots in microgravity. Collectively, our microarray dataset represents a rich source for mining novel genes that control root growth and orientation during spaceflight. |
| **Bibliography Type:** | Description: (Last Updated: 01/13/2026) |
| **Abstracts for Journals and Proceedings** | Tang Y, Sparks JA, Allen SN, Nakashima J, Blancaflor EB. "Biological Research in Canisters (BRIC 16): Actin-Dependent Gene Regulatory Networks that Modulate Root Growth and Orientation During Spaceflight." Presented at the 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010.<br>Program and abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, p. 37, 2010. , Nov-2010 |

| | |
|---|---|
| **Abstracts for Journals and Proceedings** | Nakashima J, Sparks JA, Tang Y, Blancaflor EB. "Biological Research in Canisters (BRIC-16): Root Orientation in Microgravity is Modulated by the Actin Cytoskeleton." Presented at the 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010.<br>Program and abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, p. 37, 2010. , Nov-2010 |
| **Abstracts for Journals and Proceedings** | Sparks JA, Quan L, Blancaflor EB. "Genetic and Cell Biological Studies of the hypersensitive to latrunculin B 3 (hlb3) Mutant in Arabidopsis thaliana." Presented at the 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010.<br>Program and abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, p. 36, 2010. , Nov-2010 |
| **Abstracts for Journals and Proceedings** | Gravelle A, Nakashima J, Palanichelvam K, Blancaflor EB. "Brassinolides Promote the Gravitropic Response of Maize Roots by Disrupting Actin Filament Organization." Presented at the 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010.<br>Program and abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, p. 34, 2010. , Nov-2010 |
| **Awards** | Gravelle A, Nakashima J, Palanichelvam K, Blancaflor EB. "3rd place Undergraduate Student Poster Award for: Gravelle A, Nakashima J, Palanichelvam K, Blancaflor EB. 'Brassinolides Promote the Gravitropic Response of Maize Roots by Disrupting Actin Filament Organization.' American Society for Gravitational and Space Biology 26th annual meeting, National Harbor, Maryland, November 2010." Nov-2010 |
| **Significant Media Coverage** | Calaway JA. "Boldly going. Noble Foundation researchers earn a rare opportunity to launch an experiment into space." Legacy: A Publication of The Samuel Roberts Noble Foundation. 2010 Fall;16-23.<br>http://www.noble.org/Global/news/legacy/fall2010/Legacy_Fall10.pdf , Sep-2010 |
| **Significant Media Coverage** | Chen C. "Shuttle Discovery Launches with Oklahoma Research Inside." TV Media outlet: News 9 Oklahoma City, Oklahoma, April 2010., Apr-2010 |
| **Significant Media Coverage** | Saldana M. "Noble Foundation Plants Set to Lift Off on Discovery. " TV Media Outlet: Channel 10 KTEN news, February 2010., Feb-2010 |
| **Significant Media Coverage** | Downs D. "Local scientist sends plants into space. " TV Media Outlet: First 12 news, KXII, February 2010., Feb-2010 |
| **Significant Media Coverage** | Levins S. "Plants from Ardmore now in outer space. " TV Media outlet: First news 12, KXII, April 2010., Apr-2010 |
| **Significant Media Coverage** | Saldana M. "Ardmore Plants Lift Off on Space Shuttle Discovery. " TV media outlet: KTEN news, Oklahoma, April 2010., Apr-2010 |
| **Significant Media Coverage** | Painter B. "Oklahoma researchers explore out-of-this-world agriculture. " Newspaper article: Tulsa World, November 2010., Nov-2010 |
| **Significant Media Coverage** | Painter B. "Oklahoma researchers explore out-of-this-world agriculture. " Newspaper article: The Daily Oklahoman, November 2010., Nov-2010 |
| **Significant Media Coverage** | Banker P. "Researcher discusses space experiment." Newspaper article: The Daily Ardmoreite, September 2010., Sep-2010 |
| **Significant Media Coverage** | Banker P. "Out of This World. Noble Foundation scientist sends plant to space." Newspaper article: The Daily Ardmoreite, February 2010., Feb-2010 |